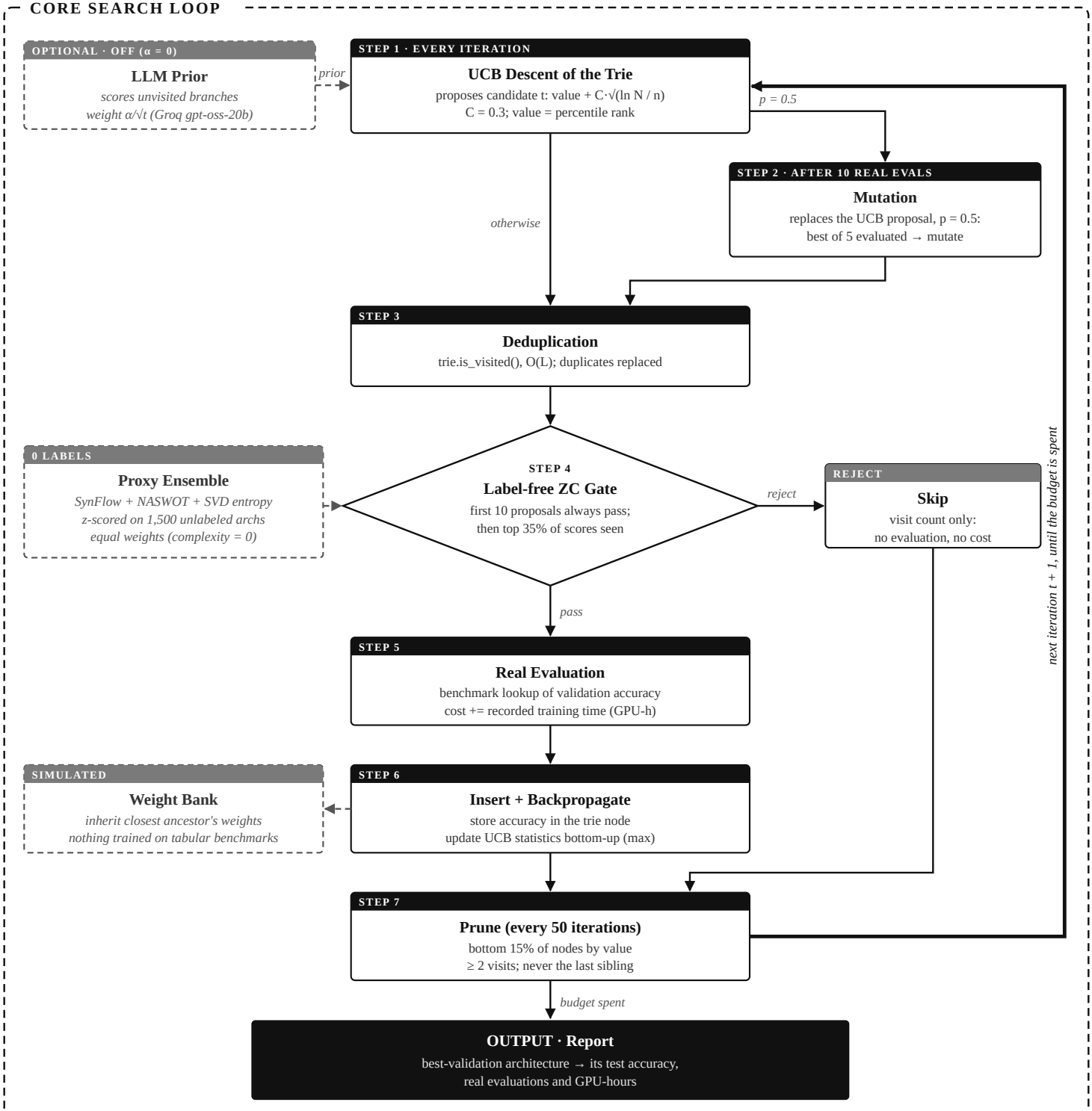

CORE SEARCH LOOP
OPTIONAL · OFF (α = 0)
LLM Prior
scores unvisited branches
weight α/√t (Groq gpt-oss-20b)
prior
STEP 1 · EVERY ITERATION
UCB Descent of the Trie
proposes candidate t: value + C·√(ln N / n)
C = 0.3; value = percentile rank
p = 0.5
STEP 2 · AFTER 10 REAL EVALS
Mutation
replaces the UCB proposal, p = 0.5:
best of 5 evaluated → mutate
otherwise
STEP 3
Deduplication
trie.is_visited(), O(L); duplicates replaced
0 LABELS
Proxy Ensemble
SynFlow + NASWOT + SVD entropy
z-scored on 1,500 unlabeled archs
equal weights (complexity = 0)
STEP 4
Label-free ZC Gate
first 10 proposals always pass;
then top 35% of scores seen
reject
REJECT
Skip
visit count only:
no evaluation, no cost
pass
STEP 5
Real Evaluation
benchmark lookup of validation accuracy
cost += recorded training time (GPU-h)
SIMULATED
Weight Bank
inherit closest ancestor's weights
nothing trained on tabular benchmarks
STEP 6
Insert + Backpropagate
store accuracy in the trie node
update UCB statistics bottom-up (max)
STEP 7
Prune (every 50 iterations)
bottom 15% of nodes by value
≥ 2 visits; never the last sibling
next iteration t + 1, until the budget is spent
budget spent
OUTPUT · Report
best-validation architecture → its test accuracy,
real evaluations and GPU-hours